



AI-Powered Live Streaming eCommerce Integration

AI-Powered Live Cooking Video Analysis

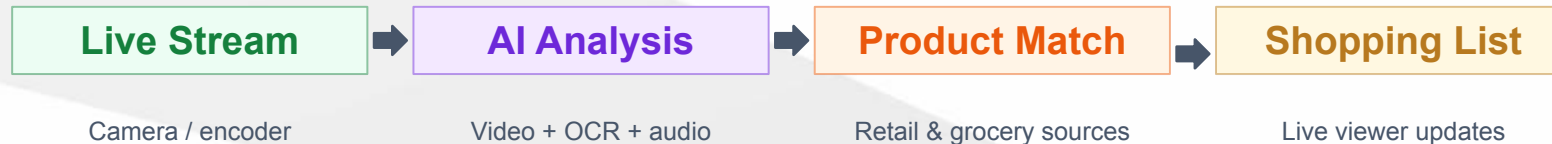
Real-time shopping list platform on Google Cloud



Executive Overview

Convert a live cooking show into an interactive shopping experience

The platform analyzes the live stream, detects ingredients and utensils, matches them to products, and updates a viewer-facing shopping list every few seconds.



Google Cloud native • scalable • secure

Business value

New monetization

Turn cooking content into shoppable moments.

Better engagement

Keep viewers active while the show is live.

Faster conversion

Reduce steps from recipe discovery to purchase.

Executive Summary

What the platform delivers for the business and end users



Core outcomes

- Analyzes live cooking video streams in near real time.
- Detects utensils, ingredients, furniture, on-screen text, and spoken recipe steps.
- Continuously updates a live shopping list during the show.
- Matches detected items to purchasable products and grocery options.
- Provides a scalable, secure, cloud-native solution on Google Cloud.

Executive lens

Revenue enablement

Sponsored products, affiliate links, and grocery/cart integrations.

Customer experience

A frictionless path from “watching” to “buying”.

Operational scalability

Managed services handle live traffic, processing bursts, and analytics.

Data intelligence

Trend insights on recipes, products, brands, and shopper interest.

Near real-time shopping list updates

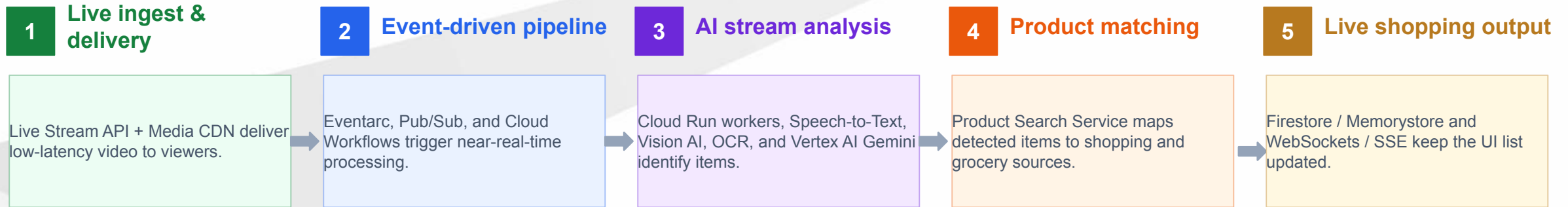
AI-driven item detection

Product discovery integration

Analytics-ready

Technical Solution Details

Google Cloud services supporting live stream ingestion, AI analysis, product matching, and real-time UI updates



Managed data layer

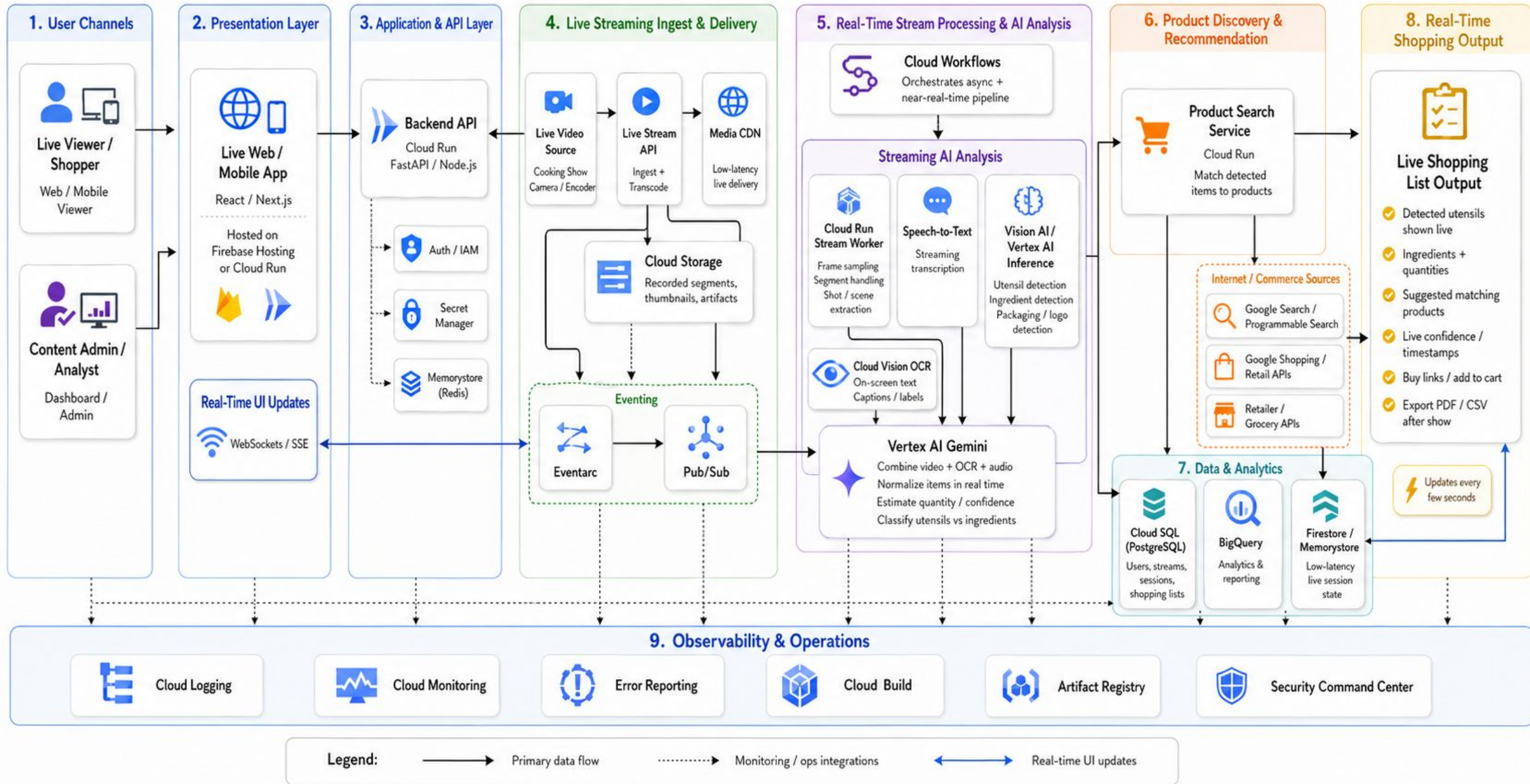
- Cloud SQL stores users, streams, sessions, carts, and shopping lists.
- Firestore / Memorystore maintains low-latency live session state.
- BigQuery supports analytics, reporting, and trend discovery.

Security & operations

- IAM and Secret Manager protect service access and external API credentials.
- Cloud Logging, Monitoring, Error Reporting, and Security Command Center support operations.
- Cloud Build and Artifact Registry enable controlled releases.

AI-Powered Live Cooking Video Analysis & Real-Time Shopping List Platform

Google Cloud Real-Time Streaming Technical Architecture





THANKS